Evaluation runs: **naive** (s1-naive)

generated 2026-11-05 from brain-data a1150b5a2 2026-11-05 14:21:32 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): naive: expected violation; self: expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	2	2	conforms	0.0	4	1 h, 250 GB	2	3	brain03, 2026-11-02
3	6	6	conforms	0.0	4	1 h, 250 GB	6	21	brain03, 2026-11-02
4	24	24	conforms	0.0	2	1 h, 250 GB	24	300	brain03, 2026-11-02
5	120	120	conforms	0.3	4	1 h, 250 GB	120	7260	brain03, 2026-11-02
6	720	720	conforms	9.4	4	1 h, 250 GB	720	259560	brain03, 2026-11-02
7	5040	5040	conforms	497.8	11	1 h, 250 GB	5040	12703320	brain03, 2026-11-02

stateless: pair **naive**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	2	2	2	violation	0.0	4	1 h, 250 GB	2	4	brain03, 2026-11-02
3	6	6	6	violation	0.0	4	1 h, 250 GB	6	36	brain03, 2026-11-02
4	24	24	24	violation	0.0	4	1 h, 250 GB	24	576	brain03, 2026-11-02
5	120	120	120	violation	0.5	2	1 h, 250 GB	120	14400	brain03, 2026-11-02
6	720	720	720	violation	18.4	9	1 h, 250 GB	720	518400	brain03, 2026-11-02
7	5040	5040	5040	violation	981.9	26	1 h, 250 GB	5040	25401600	brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, self 8 2 bag, naive 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, naive 10 1 bag, self 10 1 bag, naive 10 2 bag, self 10 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, self 4 2 bag, naive 4 2 bag, self 5 2 bag, naive 5 2 bag, naive 6 2 bag, self 6 2 bag, self 7 2 bag, naive 7 2 bag.

left out by the table rules (invalid: foreign load): naive 10 bag, naive 11 bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	2	2	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	6	6	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5	120	120	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6	720	720	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
7	5040	5040	conforms	0.4	4	1 h, 250 GB	brain02, 2026-11-02

stateful: pair **naive**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	2	2	2	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	6	6	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
4	24	24	24	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5	120	120	120	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-02
6	720	720	720	violation	0.1	6	1 h, 250 GB	brain02, 2026-11-02
7	5040	5040	5040	violation	0.6	34	1 h, 250 GB	brain02, 2026-11-02
8	40320	40320	40320	violation	5.1	267	1 h, 250 GB	brain02, 2026-11-02
9	362880	362880	362880	violation	53.7	3211	1 h, 250 GB	brain02, 2026-11-02
10	3628800	3628800	3628800	violation	609.5	41493	1 h, 250 GB	brain02, 2026-11-02
10	3628800	3628800	3628800	violation	664.1	41493	48 h, 250 GB	brain04, 2026-11-03
11				oom (250 GB)			48 h, 250 GB	brain01, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, self 8 2 bag, naive 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, naive 10 1 bag, self 10 1 bag, naive 10 2 bag, self 10 2 bag, self 14 2 bag, naive 14 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, self 4 2 bag, naive 4 2 bag, self 5 2 bag, naive 5 2 bag, naive 6 2 bag, self 6 2 bag, self 7 2 bag, naive 7 2 bag, naive 8 2 bag, naive 9 2 bag, naive 10 2 bag.

the tool ended with an error (exit-101): self 8 1 bag.

skipped: a smaller size had reached its limit: self 8 2 bag, self 9 1 bag, self 9 2 bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	6	6	conforms	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
4	24	24	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
5	120	120	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
6	720	720	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7	5040	5040	conforms	0.6	10	1 h, 250 GB	0.3	0.4	0.3	merge	brain01, 2026-11-03

offline: pair **naive**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2
3	6	6	6	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2
4	24	24	24	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2
5	120	120	120	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2
6	720	720	720	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2
7	5040	5040	5040	violation	0.7	60	10 min, 16 GB	0.3	0.4	0.3	in-memory	brain01, 2
10	3628800	3628800	3628800	violation	730.9	9811	48 h, 250 GB	384.9	397.6	333.3	merge	brain01, 2
11	39916800	39916800	39916800	violation	13349.0	115422	48 h, 250 GB	5218.0	5447.0	7902.0	merge	brain01, 2

Not run (offline). planned but never scheduled (grid points beyond the ladder): naive 8 1 bag, self 8 1 bag, self 8 2 bag, naive 8 2 bag, naive 9 1 bag, self 9 1 bag, self 9 2 bag, naive 9 2 bag, naive 10 1 bag, self 10 1 bag, naive 10 2 bag, self 10 2 bag.

left out by the table rules (ok): naive 2 2 bag, self 2 2 bag, naive 3 2 bag, self 3 2 bag, self 4 2 bag, naive 4 2 bag, self 5 2 bag, naive 5 2 bag, naive 6 2 bag, self 6 2 bag, self 7 2 bag, naive 7 2 bag.